

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 1: International & European Framework (EASA AIR-OPS)

Syllabus Source: CAE Oxford (pp. 1-36) | Pass Target: >= 90% AviationExam

1. THE ICAO & EASA REGULATORY ARCHITECTURE

Commercial flight operations in Europe are governed by EASA Regulation (EU) No 965/2012 (AIR-OPS), transposing ICAO Annex 6 standards:

EASA AIR-OPS Part	Full Official Title	Scope & Regulatory Application
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Part-GEN	General Requirements	Definitions, acronyms, and common rules across all flight operations.
Part-ARO	Authority Requirements	Requirements for National Aviation Authorities (e.g. AESA) issuing certificates.
Part-ORO	Organisation Requirements	Management systems, safety management (SMS), and Air Operator Certificates (AOC).
Part-CAT	Commercial Air Transport	Mandatory operating rules for commercial airlines carrying passengers or cargo.
Part-SPA	Specific Approvals	Operations requiring special authorization: ETOPS, RVSM, LVO, PBN, Dangerous Goods.
Part-NCC / Part-NCO	Non-Commercial Operations	Complex motor-powered (NCC) and other-than-complex (NCO) private flights.

2. THE OPERATIONS MANUAL (OM PARTS A, B, C, D)

An Air Operator Certificate (AOC) holder must maintain an Operations Manual divided into 4 mandatory parts:

- OM Part A (General / Basic): Company policies, safety management, crew responsibilities, standard operating procedures (SOP), fuel policy, and flight duty limits (FTL).
- OM Part B (Aeroplane Operating Matters): Type-specific aircraft technical systems, limitations, checklist procedures, and AFM performance tables.
- OM Part C (Route and Aerodrome Instructions): Route guides, terminal charts, airport operating minima, and escape routes.
- OM Part D (Training): Training syllabi, simulator checks, and proficiency checking for flight crew and cabin crew.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): MEL Repair Interval Categories

Legal Rectification Intervals for Inoperative Items: - Category A: No standard interval. Rectification time is specifically stipulated in remarks (e.g. 24 hours, 3 flights). - Category B: Must be rectified within 3 consecutive calendar days (72 hours, excluding day of discovery). - Category C: Must be rectified within 10 consecutive calendar days. - Category D: Must be rectified within 120 consecutive calendar days. Commander Authority Rule: The pilot-in-command has final authority to refuse an aircraft dispatched with multiple MEL defects if safety is degraded.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In which part of an airline's Operations Manual (OM) are TYPE-SPECIFIC aircraft operating procedures and

- [A]: OM Part A
- [B]: OM PART B (Aeroplane Operating Matters)
- [C]: OM Part C
- [D]: OM Part D

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under EASA ORO.MLR.100, OM-B contains all aircraft type-specific data, systems operating procedures, checklists, and performance limitations.

Q2: Which EASA AIR-OPS Annex sets out mandatory rules for operators conducting COMMERCIAL AIR TRAN

- [A]: Part-NCO
- [B]: PART-CAT (Annex IV: Commercial Air Transport Operations)
- [C]: Part-SPO
- [D]: Part-MED

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Part-CAT applies to all commercial revenue flights carrying passengers, cargo, or mail.

Q3: Who holds the final legal authority for the safety and operation of the aircraft during flight?

- [A]: The airline Chief Pilot
- [B]: THE COMMANDER (Pilot-in-Command)
- [C]: Air Traffic Control
- [D]: The dispatcher

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Annex 6 and EASA CAT.GEN.MPA.105, the Commander has complete authority over the aircraft and all persons on board.

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Chapter 2: Crew Composition & Flight Duty Limitations (FTL)

Syllabus Source: CAE Oxford (pp. 37-76) | Pass Target: >= 90% AviationExam

1. EASA FLIGHT AND DUTY TIME LIMITATIONS (FTL)

EASA Subpart-FTL (ORO.FTL) establishes mandatory limits to prevent crew fatigue from compromising safety:

- Flight Duty Period (FDP): Time from when a crew member is required to report for duty (typically 45 to 60 min before departure) until engine shutdown at the end of the last sector.
- Maximum Daily FDP: Ranges from 9 to 13 hours, depending on report time (circadian rhythm) and number of flight sectors (e.g. 1 to 2 sectors reporting between 06:00 and 13:29 allows 13.0 hours FDP; reporting between 02:00 and 04:59 drops to 9.0 hours!).
- Minimum Rest Period: Mandatory rest before duty must be AT LEAST 12 HOURS, or equal to the duration of the preceding duty period (whichever is GREATER!). If away from home base: at least 10 hours including an 8-hour sleep opportunity.

2. CUMULATIVE DUTY & FLIGHT TIME CAPS

- Maximum Cumulative Duty Hours: 60 duty hours in 7 consecutive days; 110 duty hours in 14 consecutive days; 190 duty hours in 28 consecutive days.
- Maximum Cumulative Flight Hours (Block Time): 100 flight hours in any 28 consecutive days; 900 flight hours in a calendar year; 1,000 flight hours in any 12 consecutive months.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Maximum Daily FDP Calculation

Problem: An acclimatized 2-pilot crew reports at 07:00 local for 4 sectors. Find the maximum legal Flight Duty Period under EASA ORO.FTL. Step 1: Check report time in ORO.FTL table: 07:00 (Favorable bracket 06:00 to 13:29). Step 2: Check sector column for 4 sectors: - 1-2 sectors: Max FDP = 13h 00m. - 3 sectors: Max FDP = 12h 30m. - 4 sectors: Max FDP = 12h 00m! Rule of Thumb: Each additional sector beyond 2 reduces allowed FDP by 30 minutes.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Minimum Rest Period after Heavy Duty

SCENARIO (EASA Flight Crew Rostering Drill): A commercial airline crew reports for duty at home base at 06:00 local time: - They operate a multi-sector European schedule - The duty finishes at 20:30 local time (Total duty period = 14.5 hours) QUESTION: What is the MINIMUM REST PERIOD required before the crew can report for their next duty? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Recall the EASA Home Base Rest Rule: - Rule: Minimum rest at home base = The GREATER of: * (A) 12 hours standard baseline, OR * (B) The exact duration of the preceding duty period! Step 2: Compare the two values: - Standard Baseline = 12.0 hours - Preceding Duty Period = 14.5 hours - The greater value is 14.5 hours! Step 3: Calculate the Earliest Allowable Next Report Time: - Duty finished at 20:30. - Add 14.5 hours of rest: - 20:30 + 14h 30m = 11:00 the following morning! FINAL ANSWER: Minimum mandatory rest period is 14.5 hours (14 hours 30 minutes). Earliest next report is 11:00.

[!] EXAM TRAP (AVIATIONEXAM) - The Rest Away from Base vs Home Base Trap

- At HOME BASE: Rest = Greater of 12 hours OR length of duty. - AWAY FROM BASE: Rest = Greater of 10 hours OR length of duty (must include an 8-hour sleep opportunity). - Never assign 10 hours rest at home base; that is an illegal roster violation!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA FTL regulations, what is the maximum cumulative FLIGHT TIME (block hours) permitted in any 28 consecutive days?

- [A]: 60 hours
- [B]: 100 FLIGHT HOURS
- [C]: 110 hours
- [D]: 900 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. EASA ORO.FTL.210 sets maximum cumulative flight time at 100 hours in 28 consecutive days, 900 hours in a calendar year, and 1,000 hours in 12 months.

Q2: What is the maximum cumulative DUTY TIME permitted in any 7 consecutive days?

- [A]: 40 hours
- [B]: 60 DUTY HOURS
- [C]: 70 hours
- [D]: 100 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Total duty periods must not exceed 60 hours in 7 consecutive days, 110 hours in 14 days, and 190 hours in 28 days.

Q3: What is the minimum required rest period at home base following a duty period of 10 hours?

- [A]: 10 hours
- [B]: AT LEAST 12 HOURS (the baseline minimum)
- [C]: 8 hours
- [D]: 24 hours

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At home base, rest must be at least 12 hours, or the length of the preceding duty if longer. Since $10h < 12h$, the 12-hour minimum applies.

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Chapter 3: Aerodrome Operating Minima (AOM) & LVO

Syllabus Source: CAE Oxford (pp. 77-116) | Pass Target: >= 90% AviationExam

1. ALL-WEATHER PRECISION APPROACH CATEGORIES

Under EASA AIR-OPS (CAT.OP.MPA.110), aerodrome operating minima establish legal limits for take-off and landing:

ILS Approach Category	Decision Height (DH)	Runway Visual Range (RVR) Minimum
ILS Approach Category	Decision Height (DH)	Runway Visual Range (RVR) Minimum
Category I (CAT I)	DH >= 200 ft (60 m)	RVR >= 550 meters (or 800 m for manual landing without HUD).
Category II (CAT II)	DH: 100 ft to < 200 ft (30 to 60 m)	RVR >= 300 meters (requires autoland or HUD guidance).
Category IIIa (CAT IIIa)	DH: < 100 ft (or NO DH)	RVR >= 175 meters (requires fail-passive autoland).
Category IIIb (CAT IIIb)	DH: < 50 ft (or NO DH)	RVR: 75 to < 175 meters (requires fail-operational autoland and rollout).
Category IIIc (CAT IIIc)	NO Decision Height (0 ft)	NO RVR limitation (Zero visibility landing and rollout - not operational).

2. LOW VISIBILITY TAKE-OFF (LVTO)

A take-off conducted with RVR < 400 meters is an LVTO operation requiring Specific Approval (Part-SPA.LVO). Absolute minimum RVR for commercial jets: 125 meters with runway centerline lights and markings (or 150 m for Cat C/D).

3. THE APPROACH BAN RULE

The flight crew may begin an instrument approach regardless of reported RVR. However:

- The 1,000 ft Ban Point: If the reported RVR is LESS than the published minimum, THE APPROACH MUST NOT BE CONTINUED BEYOND: The Outer Marker (OM), or an altitude of 1,000 ft above aerodrome elevation in the final approach segment!
- Ban Exemption After 1,000 ft: If the aircraft is ALREADY BELOW 1,000 ft AGL and RVR deteriorates below minima, the crew MAY CONTINUE to the Decision Height (DH). If visual references are acquired at DH, landing is permitted!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): The Approach Ban Decision Tree

Step 1: Check threshold point: 1,000 ft above aerodrome level (AAL) or Final Approach Fix (FAF). Step 2: Prior to reaching 1,000 ft AAL: - If reported RVR < published minimum -> APPROACH BAN! Cannot continue past 1,000 ft; execute go-around. - If reported RVR >= published minimum -> Cleared to continue past 1,000 ft AAL. Step 3: If below 1,000 ft AAL and reported RVR subsequently drops below minimum: - Legal permission to continue to Decision Altitude (DA) or MDA! Step 4: At DA/MDA: If required visual references are acquired -> LAND. If not acquired -> GO AROUND!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA regulations, what are the Decision Height (DH) and minimum RVR limits for an ILS CATEGORY II approach?

- [A]: DH $\geq$ 200 ft, RVR $\geq$ 550 m
- [B]: DH between 100 ft and $<$ 200 ft, and RVR NOT LESS THAN 300 METERS
- [C]: DH $<$ 50 ft, RVR 75 m
- [D]: Zero DH

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Cat II precision approach is defined by a DH between 100 and 199 ft, and a minimum RVR of 300 m.

Q2: According to the 'APPROACH BAN' rule, past which point is an aircraft prohibited from continuing an approach if the RVR drops below legal minima?

- [A]: Top of descent
- [B]: The 1,000 FT AGL POINT (or Outer Marker) in the final approach segment
- [C]: Initial approach fix
- [D]: Touchdown

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If RVR drops below legal minima before passing 1,000 ft AGL (or the outer marker), continuing the approach is illegal; a missed approach must be initiated.

Q3: What special equipment is required on the runway for a commercial jet Low Visibility Take-Off (LVTO) with a minimum RVR of 400 m?

- [A]: Only edge lights
- [B]: High-intensity RUNWAY CENTERLINE LIGHTS and runway markings
- [C]: Sodium flares
- [D]: Ground radar only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. LVTO below 400 m requires illuminated runway centerline lights and markings to provide continuous lateral visual tracking.

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Chapter 4: PANS-OPS Instrument Procedures & Circling

Syllabus Source: CAE Oxford (pp. 117-156) | Pass Target: $\geq 90\%$ AviationExam

1. PANS-OPS INSTRUMENT DEPARTURE CRITERIA

ICAO Doc 8168 (PANS-OPS) establishes standard criteria for protected airspace design:

- Design Climb Gradient (PDG): Standard departure climb gradient is 3.3% (200 ft/NM), comprising 2.5% net obstacle clearance + 0.8% safety buffer.
- Minimum Turn Altitude: No turn may be initiated on departure below 120 METERS (400 FT) ABOVE AERODROME ELEVATION!

2. VISUAL MANOEUVRING (CIRCLING APPROACH)

A visual circling approach is used when landing runway does not align with the instrument approach (alignment $> 30^\circ$ off runway centerline, or descent gradient exceeds 6.5%):

- Protected Circling Radii: Cat A = 1.68 NM; Cat B = 2.66 NM; Cat C = 4.20 NM; Cat D = 5.28 NM from all runway thresholds.
- Action on Loss of Visual Reference: IF VISUAL CONTACT WITH THE RUNWAY IS LOST DURING CIRCLING: Initiate an IMMEDIATE CLIMBING TURN TOWARDS THE LANDING RUNWAY, establish overhead the aerodrome, and follow the missed approach procedure of the INSTRUMENT RUNWAY originally flown!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Circling Missed Approach Procedure

Golden Rule: If visual reference is lost at any point during visual circling to land: Step 1: Apply immediate TOGA thrust and initiate positive climb. Step 2: Turn TOWARDS the landing runway centerline to stay within protected obstacle clearance area. Step 3: Establish the aircraft on the published missed approach procedure of the initial instrument approach! Fatal Trap: Never turn away from the runway into unprotected airspace during a circling go-around.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under PANS-OPS design rules, what is the MINIMUM altitude above aerodrome level before an aircraft may

- [A]: 200 ft
- [B]: 120 METERS (400 FEET) AGL
- [C]: 1,000 ft
- [D]: 500 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. PANS-OPS prohibits turns below 120 m (400 ft) above aerodrome elevation to guarantee obstacle clearance during straight acceleration.

Q2: If visual reference is LOST while conducting a visual circling manoeuvre to land, what is the mandatory pil

- [A]: Descend 500 ft to find the ground
- [B]: Initiate an IMMEDIATE CLIMBING TURN TOWARDS THE LANDING RUNWAY, climb overhead, and fly the missed approach of the instrument approach runway
- [C]: Turn 90° away from airport
- [D]: Land immediately

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Turning towards the runway keeps the aircraft within the protected circling obstacle area while climbing to safety in the instrument missed approach.

Q3: What is the radius of the protected obstacle clearance area for a CATEGORY C aircraft conducting a circling approach?

- [A]: 1.68 NM
- [B]: 4.20 NAUTICAL MILES from the runway threshold
- [C]: 2.66 NM
- [D]: 5.28 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. PANS-OPS assigns Cat C aircraft a circling area radius of 4.20 NM based on a maximum circling speed of 180 kt.

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Chapter 5: Extended Diversion Time Operations (ETOPS / EDTO)

Syllabus Source: CAE Oxford (pp. 157-196) | Pass Target: $\geq 90\%$ AviationExam

1. THE ETOPS REGULATORY CONCEPT

ETOPS (Extended-range Twin-engine Operational Performance Standards) / EDTO permits twin-engine aircraft to fly long-range routes across oceans and deserts:

- The 60-Minute Threshold: Without ETOPS approval, a twin-jet must remain within 60 MINUTES FLYING TIME at One-Engine-Inoperative (OEI) cruise speed in still air from an adequate aerodrome.
- ETOPS Approval Ratings: ETOPS 120 min, 180 min, or 240+ min. The approved diversion time defines a circle around en-route alternates: $\text{Radius (NM)} = \text{Approved ETOPS Time (hours)} \times \text{Approved OEI Cruise TAS (knots)}$!

2. ADEQUATE VS SUITABLE AERODROME

- Adequate Aerodrome: Physically capable: Runway length, strength, emergency services (RFFS), and lighting meet aircraft requirements.
- Suitable Aerodrome (ETOPS En-Route Alternate): An adequate aerodrome whose FORECAST WEATHER AT EXPECTED TIME OF ARRIVAL (ETA) is AT OR ABOVE ETOPS DISPATCH MINIMA (which includes safety buffers above normal landing minima)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): ETOPS Threshold Distance & Altitude

Problem: A twin-jet has an approved single-engine cruise speed of 420 kt TAS in standard atmosphere. Find the threshold distance beyond which ETOPS regulations apply. Step 1: ETOPS threshold is 60 minutes (1.0 hour) single-engine cruise speed. Step 2: Threshold Distance = $420 \text{ kt} \times 1.0 \text{ h} = 420 \text{ NM}$! Step 3: If any point on the route is $> 420 \text{ NM}$ from an adequate airport, ETOPS approval is mandatory. Step 4: If operator holds ETOPS 180 min approval: Maximum allowable diversion distance is $420 \text{ kt} \times 3.0 \text{ h} = 1,260 \text{ NM}$.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating ETOPS 120-Minute Maximum Diversion Range

SCENARIO (Transatlantic Route Planning Drill): An Airbus A330 operator has ETOPS 120-minute approval: - Approved One-Engine Inoperative (OEI) cruise speed in still air = 420 knots TAS QUESTION: What is the maximum allowable diversion distance (radius of ETOPS circle) in Nautical Miles from an en-route alternate aerodrome? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify Approved Diversion Time and OEI Speed: - ETOPS Time = 120 minutes = 2.0 hours. - Approved OEI Speed = 420 knots TAS. Step 2: Recall the ETOPS Maximum Diversion Distance Formula: - Rule: ETOPS distance is calculated strictly in STILL AIR (zero wind credit/penalty)! - Formula: Maximum Range (NM) = Diversion Time (hours) x OEI TAS Step 3: Calculate the Maximum Range: - Range = 2.0 hours x 420 knots = 840 Nautical Miles! Step 4: Operational Meaning for Dummies: - The aircraft must never be further than 840 NM from a suitable ETOPS alternate at any point along the ocean route! FINAL ANSWER: Maximum allowable diversion distance is exactly 840 Nautical Miles.

[!] EXAM TRAP (AVIATIONEXAM) - The Still Air ETOPS Distance Trap

- The maximum ETOPS diversion distance is calculated in STILL AIR (zero wind)! - Even if there is a 100 knot jet stream tailwind, you CANNOT stretch the 840 NM circle! - The circle radius is fixed by: Approved Hours x OEI TAS in still air.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the baseline threshold diversion time for twin-turbine aeroplanes beyond which ETOPS / EDTO approval is required?

- [A]: 30 minutes
- [B]: 60 MINUTES at one-engine-inoperative cruise speed in still air
- [C]: 120 minutes
- [D]: 180 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Any twin-engine commercial flight route that exceeds 60 minutes diversion time to an adequate aerodrome is classified as an extended diversion time operation (ETOPS).

Q2: In ETOPS dispatch regulations, what distinguishes a 'SUITABLE AERODROME' from an 'ADEQUATE AERODROME'?

- [A]: Only runway length
- [B]: A suitable aerodrome must have forecast weather at ETA meeting or exceeding higher ETOPS DISPATCH MINIMA
- [C]: Suitable aerodromes must be military
- [D]: There is no difference

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An adequate aerodrome meets physical criteria; a suitable aerodrome also has confirmed weather forecasts meeting strict ETOPS dispatch buffers during the arrival window.

Q3: An operator has 180-minute ETOPS approval. If their approved OEI cruise speed is 400 kt TAS in still air, what is the maximum allowable diversion distance (radius of ETOPS circle) in Nautical Miles?

- [A]: 800 NM
- [B]: 1,200 NAUTICAL MILES (3.0 hours x 400 kt = 1,200 NM)

- [C]: 400 NM
- [D]: 1,800 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. 180 minutes = 3 hours. 3 hours x 400 knots = 1,200 NM maximum distance from an en-route alternate.

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Chapter 6: Special Approvals: RVSM, NAT HLA & Contingencies

Syllabus Source: CAE Oxford (pp. 197-234) | Pass Target: >= 90% AviationExam

1. REDUCED VERTICAL SEPARATION MINIMUM (RVSM)

RVSM reduces vertical separation between aircraft from 2,000 ft to 1,000 ft between FL 290 and FL 410 inclusive, doubling airspace capacity:

- **Mandatory RVSM Equipment (Part-SPA.RVSM):** 1. Two independent primary altimetry systems (maximum difference ± 200 ft in flight); 2. One automatic altitude-control system (autopilot holding altitude within ± 65 ft); 3. One altitude-alerting system (± 300 ft alert threshold); 4. One Mode C / Mode S altitude-reporting transponder.
- **Strategic Lateral Offset Procedures (SLOP):** In oceanic airspace (NAT HLA), pilots are authorized to fly 1 NM or 2 NM RIGHT OF TRACK to disperse wake turbulence and reduce collision risk without notifying ATC!

2. OCEANIC IN-FLIGHT CONTINGENCY PROCEDURE (NAT HLA)

If unable to maintain assigned flight level or route in NAT HLA due to engine failure, depressurization, or medical emergency:

- **Step 1: Turn:** Turn 30° left or right of track (away from nearest route/traffic).
- **Step 2: Offset:** Fly to acquire and maintain a 5 NAUTICAL MILE lateral offset from the track centerline.
- **Step 3: Level Change:** Once established on the 5 NM offset, climb or descend: By 500 FT (if below FL 410); By 1,000 FT (if at or above FL 410) to avoid standard traffic levels!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): NAT HLA Oceanic In-Flight Contingency**

Emergency Descent / Engine Failure without ATC Clearance in North Atlantic HLA: Step 1: Turn on all exterior lights and squawk 7700. Step 2: Turn 30 degrees left or right off track (away from adjacent organized tracks). Step 3: Fly until establishing a 5 NM parallel offset from original track. Step 4: Once established on the 5 NM offset: - If remaining in RVSM airspace (FL 290 - FL 410): Climb or descend by 300 ft to fly an intermediate flight level (e.g. FL 353). - If descending below FL 290: Descend to suitable altitude. Step 5: Broadcast MAYDAY on 121.5 MHz and air-to-air 123.45 MHz.

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Executing Oceanic Engine-Out Contingency

SCENARIO (Transatlantic Emergency Descent Drill): A twin-jet airliner is cruising at FL 370 in the North Atlantic (NAT HLA) on Track Charlie (Track 090° True): - An engine fails catastrophically! - Unable to maintain FL 370, the crew cannot establish radio communication with Shanwick ATC immediately QUESTION: What precise maneuver must the crew execute under ICAO NAT Doc 007 contingency rules? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Initial Action: - Select Maximum Continuous Thrust on good engine, set transponder to 7700, turn on all exterior lights! Step 2: Initiate Lateral Turn: - Turn 30° to the RIGHT (heading 120°) or LEFT (heading 060°) away from conflicting tracks. Step 3: Establish 5 NM Lateral Offset: - Fly out until 5.0 NM away from the centerline of Track Charlie. - Turn back onto track heading (090° True) to parallel the track 5 NM displaced. Step 4: Select Contingency Level (Divert from FL 370): - Normal traffic flies at whole flight levels: FL 370, FL 360, FL 350, FL 340. - To avoid hitting opposing or same-direction traffic below: - Aircraft must descend to an INTERMEDIATE 500 FT LEVEL! - Drift down to: FL 335, FL 315, or FL 295 (500 ft offset from normal traffic levels)! FINAL ANSWER: Turn 30° to establish 5 NM offset, then descend to a 500-ft offset flight level (e.g. FL 335 / FL 315).

[!] EXAM TRAP (AVIATIONEXAM) - The 500 ft vs 1,000 ft Contingency Offset Trap

- If below FL 410: The altitude offset is 500 FT (e.g. FL 335). - If at or above FL 410: The altitude offset is 1,000 FT. - The lateral offset is ALWAYS 5 NM in modern ICAO oceanic airspace (changed from the old 15 NM rule).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: In RVSM airspace (FL 290 to FL 410), what is the maximum permissible difference between the two primary**

- [A]: 50 ft
- [B]: NOT EXCEEDING 200 FEET
- [C]: 300 ft
- [D]: 500 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under EASA SPA.RVSM.110, the maximum difference between primary altimeters in level flight is 200 ft (and within 75 ft on the ground).

Q2: Under Strategic Lateral Offset Procedures (SLOP) in oceanic airspace, what lateral offsets are authorized t

- [A]: 10 NM right
- [B]: 1 NM or 2 NM RIGHT OF TRACK (or tenths of a mile up to 2 NM)
- [C]: 5 NM left
- [D]: 1 NM left

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. SLOP allows pilots to offset 1 or 2 NM right of centerline to disperse wake turbulence and minimize collision risk.

Q3: What is the lateral offset distance required when executing an oceanic in-flight contingency in the North A

- [A]: 15 NM
- [B]: Exactly 5 NAUTICAL MILES

- [C]: 30 NM
- [D]: 1 NM

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under modern ICAO Doc 007 procedures, the oceanic contingency offset is standardized at 5 NM.

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Chapter 7: In-Flight Emergencies: Descent, Evacuation & Ditching

Syllabus Source: CAE Oxford (pp. 235-272) | Pass Target: >= 90% AviationExam

1. EMERGENCY DESCENT (RAPID DEPRESSURIZATION)

A sudden cabin decompression at high altitude (e.g. FL 390) demands immediate flight crew action before Time of Useful Consciousness (TUC < 30 seconds) expires:

- Immediate Memory Items: 1. DON OXYGEN MASKS 100% / Emergency; 2. Establish crew communications; 3. Disengage autothrottle and retard thrust levers to IDLE; 4. EXTEND SPEED BRAKES fully; 5. Pitch down into emergency descent at Vmo / Mmo; 6. Set transponder 7700.
- Target Level: Descend rapidly to 10,000 ft MSL (or FL 100), where oxygen is breathable without supplemental supply, or to the Minimum Sector Altitude (MSA) if flying over mountainous terrain!

2. EMERGENCY EVACUATION STANDARDS

Under CS-25 certification, an airliner must demonstrate complete evacuation of maximum passenger capacity within 90 SECONDS with 50% OF EXITS BLOCKED, using emergency floor lighting and inflatable escape slides.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Emergency Descent Memory Item

Rapid Depressurization Immediate Actions: 1. Oxygen masks: ON 100% in under 5 seconds, establish crew communications. 2. Pilot Flying (PF): - Thrust levers: IDLE. - Speedbrakes: FULL. - Pitch down to Vmo / Mmo. - Turn 90 degrees off airway to avoid colliding with traffic below. 3. Pilot Monitoring (PM): Squawk 7700, turn on lights, declare MAYDAY to ATC. 4. Level off at 10,000 ft or Minimum Sector Altitude (MSA), whichever is higher.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In the event of a rapid decompression at FL 370, what is the VERY FIRST memory action the pilot flying must take?

- [A]: Declare a Mayday to ATC
- [B]: DON OXYGEN MASK (100% / Emergency) and establish crew communications
- [C]: Extend speed brakes
- [D]: Retard thrust levers

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At FL 370, Time of Useful Consciousness is less than 30 seconds. Immediate donning of positive-pressure oxygen masks is paramount to prevent pilot incapacitation.

Q2: Under CS-25 airworthiness regulations, within what maximum time must an airliner demonstrate full passenger evacuation?

- [A]: 60 seconds
- [B]: 90 SECONDS with 50% of available emergency exits blocked
- [C]: 120 seconds
- [D]: 3 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Evacuation demonstration requires all passengers and crew to evacuate in dark conditions within 90 seconds using only half the exits.

Q3: When ditching an aircraft into open ocean waters with significant swell, in what direction relative to the sea swell should the aircraft be ditched?

- [A]: Directly into the face of the highest swell
- [B]: PARALLEL TO THE SWELL CRESTS (along the swell), touching down along the top of a swell or along the back of a swell
- [C]: Downwind
- [D]: Perpendicular to waves

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Landing parallel to the swell prevents the aircraft burying its nose into the oncoming wall of water, minimizing deceleration forces.

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Chapter 8: All-Weather Hazards: Windshear, Ash & Wake Turbulence

Syllabus Source: CAE Oxford (pp. 273-310) | Pass Target: >= 90% AviationExam

1. WINDSHEAR RECOGNITION & ESCAPE MANEUVER

Low-level windshear produces sudden changes in indicated airspeed (> 15 kt), vertical speed (> 500 ft/min), and pitch angle (> 5°):

- Immediate Escape Maneuver: 1. Disconnect autopilot; 2. Advance thrust levers to FULL TOGA (firewall power!); 3. Rotate wings-level towards the pitch limit indicator (stick shaker); 4. STRICTLY PROHIBITED: DO NOT CHANGE GEAR OR FLAP CONFIGURATION until terrain clearance is assured (changing gear causes massive temporary drag increase!).

2. ICAO WAKE TURBULENCE SEPARATION MINIMA

Wake vortices trail from wingtips, descending at 400 to 500 ft/min. Maximum hazard occurs behind Heavy aircraft in clean, slow, high-angle-of-attack flight:

Leading Aircraft Category	Following Aircraft Category	Departure Time Separation (Same Runway / Intersection)
Leading Aircraft Category	Following Aircraft Category	Departure Time Separation (Same Runway / Intersection)
HEAVY (>= 136,000 kg)	MEDIUM (7,000 to 136,000 kg)	2 MINUTES (3 minutes if following from intermediate intersection).
HEAVY (>= 136,000 kg)	LIGHT (< 7,000 kg)	2 MINUTES (3 minutes if following from intermediate intersection).
SUPER (Airbus A380)	HEAVY / MEDIUM	3 MINUTES full runway (4 minutes from intersection).
SUPER (Airbus A380)	LIGHT	4 MINUTES (5 minutes from intersection).

3. VOLCANIC ASH CLOUD ENCOUNTER PROCEDURES

- Engine Flameout Threat: Volcanic ash melts at 1,100°C inside turbine combustors, fusing into glassy enamel that suffocates turbine cooling holes and causes ALL ENGINES TO FLAMEOUT!
- Escape Actions: 1. Execute IMMEDIATE 180° TURN back out of ash cloud; 2. Retard thrust levers to IDLE (reduces combustion temperature below ash melting point); 3. Start APU; 4. Turn on continuous engine ignition; 5. Select maximum cabin air bleed.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Wake Turbulence Avoidance Rule

Takeoff Behind a Heavy Jet: - Lift-off occurs at rotation (V_r). Lift-off BEFORE the preceding aircraft rotation point and climb above its flight path. - Minimum ATC separation: 2 minutes (3 minutes from intermediate intersection). Landing Behind a Heavy Jet: - Stay ABOVE the preceding aircraft glidepath. - Touch down BEYOND the preceding nose wheel touchdown point (vortices cease upon nose wheel contact).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: During an in-flight encounter with low-level WINDSHEAR, why is changing flap or landing gear configuration

- [A]: Flap motor will overheat
- [B]: Retracting or extending gear/flaps creates significant TRANSIENT DRAG and loss of lift, drastically increasing the risk of ground impact
- [C]: ATC bans gear changes
- [D]: Altimeter will fail

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The transient drag during gear door cycling can cause immediate loss of the precious few feet of altitude remaining.

Q2: What is the mandatory ICAO wake turbulence departure time interval between a HEAVY aircraft taking off a

- [A]: 1 minute
- [B]: AT LEAST 2 MINUTES
- [C]: 5 minutes
- [D]: 30 seconds

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. ICAO Doc 4444 requires a minimum 2-minute separation behind a Heavy aircraft for same-runway departures (increased to 3 minutes for intersection take-offs).

Q3: When inadvertently penetrating a volcanic ash cloud, why should engine thrust levers be retarded to IDLE

- [A]: To save fuel
- [B]: To LOWER TURBINE GAS TEMPERATURES below the melting point of volcanic ash (~1,100°C), preventing molten glass from accumulating on turbine blades
- [C]: To increase airspeed
- [D]: To pressurize cabin

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At idle thrust, EGT drops below the ~1,100°C melting point of silicate ash, stopping the buildup of molten glassy deposits on turbine guide vanes.

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Chapter 9: Ground De-Icing, Anti-Icing & Holdover Times (HOT)

Syllabus Source: CAE Oxford (pp. 311-346) | Pass Target: >= 90% AviationExam

1. THE CLEAN AIRCRAFT CONCEPT & STANDARD FLUIDS

Under EASA CAT.OP.MPA.250, no aircraft may commence take-off unless all critical surfaces (wings, control surfaces, engine inlets, pitot tubes) are completely free of frost, ice, slush, or snow:

ISO / SAE Fluid Type	Viscosity, Thickener & Dye Color	Aerodynamic & Operational Characteristics
ISO / SAE Fluid Type	Viscosity, Thickener & Dye Color	Aerodynamic & Operational Characteristics
Type I (De-icing)	Unthickened, low viscosity. DYED ORANGE.	Applied hot (60°C) to remove ice/snow. Minimal Holdover Time (HOT: 5 to 15 min).
Type II (Anti-icing)	Thickened pseudoplastic. DYED YELLOW / CLEAR.	Contains polymeric thickeners that absorb freezing moisture. Minimum rotation speed: 100 kt. HOT: 30 to 45 min.
Type III (Commuter)	Thickened, intermediate viscosity. DYED BRIGHT GREEN.	Designed for commuter turboprops with low rotation speed (Vr < 100 kt).
Type IV (Advanced Anti-icing)	Highly thickened pseudoplastic. DYED EMERALD GREEN.	Provides LONGEST HOLDOVER TIME. Shears off cleanly during take-off roll at Vr >= 100 kt.

2. HOLDOVER TIME (HOT) RULES

Holdover Time is the estimated time anti-icing fluid will prevent frost/ice forming on treated aircraft surfaces:

- Exact Start of HOT: Holdover Time BEGINS AT THE EXACT MOMENT THE FINAL ANTI-ICING APPLICATION COMMENCES! (Not when treatment finishes!).
- Pre-Take-Off Contamination Check: If HOT expires prior to take-off, the aircraft must return for complete de-icing/anti-icing re-treatment, unless an external pre-take-off contamination inspection confirms surfaces remain 100% clean.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Holdover Time (HOT) Management

Clean Aircraft Concept Rules: Step 1: When final Type IV anti-icing fluid application begins: - START STOPWATCH! Holdover Time begins at START of final application, NOT when completed. Step 2: Check HOT table for fluid type, outside temperature, and precipitation type (snow, freezing fog, rain). Step 3: If taxi delay approaches end of HOT: - Inside HOT: Surfaces glossy and fluid -> Cleared for takeoff. - Expired HOT: Perform pre-takeoff contamination check. If surfaces lose gloss or show snow adhesion -> RETURN FOR RE-DE-ICING!

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Exactly when does the HOLDOVER TIME (HOT) of an anti-icing fluid treatment begin?

- [A]: When the de-icing truck leaves the aircraft
- [B]: At the START of the final anti-icing application
- [C]: When take-off clearance is received
- [D]: At brake release

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. By international ICAO and EASA definition, holdover time begins at the exact start of the final anti-icing application step.

Q2: What visual color dye is standard for SAE TYPE IV advanced anti-icing fluid?

- [A]: Orange
- [B]: EMERALD GREEN
- [C]: Yellow
- [D]: Blue

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Type I is orange; Type II is straw/yellow; Type III is bright green; Type IV is emerald green.

Q3: What happens to a thickened pseudoplastic anti-icing fluid (Type II / IV) during the aircraft's take-off ground roll?

- [A]: It freezes solid
- [B]: Under aerodynamic shear stress at speeds above 100 kt, its viscosity collapses and it SHEARS COMPLETELY OFF THE WING, leaving a clean aerodynamic surface
- [C]: It remains permanently on the wing
- [D]: It dissolves fuel

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Pseudoplastic fluids thin under shear stress, flowing completely off the wing at high speed during rotation to avoid reducing wing lift.

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Chapter 10: Carriage of Dangerous Goods by Air (ICAO & IATA)

Syllabus Source: CAE Oxford (pp. 347-380) | Pass Target: >= 90% AviationExam

1. THE 9 CLASSES OF DANGEROUS GOODS

Under ICAO Annex 18 and the IATA Dangerous Goods Regulations (DGR), hazardous materials are categorized into 9 distinct hazard classes:

Class	Dangerous Goods Category	Examples & Aircraft Loading Restrictions
Class	Dangerous Goods Category	Examples & Aircraft Loading Restrictions
Class 1	Explosives (Divisions 1.1 to 1.6)	Ammunition, flares, fireworks. Most are forbidden on passenger aircraft.
Class 2	Gases (Flammable, Non-flammable, Toxic)	Aerosols, oxygen cylinders, compressed butane, toxic chlorine.
Class 3	Flammable Liquids	Paints, aviation fuels, alcohol solvents (flash point <= 60°C).
Class 4	Flammable Solids	Matches, white phosphorus, magnesium alloys, spontaneously combustible goods.
Class 5	Oxidizing Substances & Organic Peroxides	Chemical oxygen generators (ValuJet tragedy), hydrogen peroxide.
Class 6	Toxic & Infectious Substances	Pesticides, cyanide, medical viral samples (Division 6.2).
Class 7	Radioactive Material	Medical isotopes, industrial radiotracers. Classified into Category I, II, III.
Class 8	Corrosives	Sulfuric acid, battery fluids, mercury (corrodes aluminum alloy airframes!).
Class 9	Miscellaneous Dangerous Goods	LITHIUM-ION BATTERIES, dry ice (solid CO2 creates asphyxiation risk in holds).

2. NOTIFICATION TO CAPTAIN (NOTOC)

Before departure, the airline ground handling agent must deliver a written NOTOC document to the aircraft Commander:

- Mandatory NOTOC Data: 1. Proper Shipping Name and UN Number; 2. Hazard Class and Division; 3. Net quantity per package; 4. Exact physical cargo hold compartment location (e.g. Hold 4, Bin 2); 5. Confirmation that packages are undamaged; 6. ICAO Emergency Response Drill Code (e.g. Drill 4L for emergency checklists).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): NOTOC Verification & In-Flight D

Notification to Captain (NOTOC) Verification: Step 1: Confirm written NOTOC is delivered before door closure: verify UN numbers, class, quantity, and exact cargo hold/bay. Step 2: Check ICAO Emergency Drill Code on NOTOC (e.g. 4L). Step 3: Cargo Fire / Spillage Action: - Discharge cargo fire extinguisher bottle 1 immediately (Halon). - Confirm slow-rate discharge bottle 2 initiates. - Divert to nearest suitable airport, declare MAYDAY, and pass UN numbers and drill codes to ATC.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the formal document that MUST be provided in writing to the aircraft Commander prior to departure?**

- [A]: Flight manifest
- [B]: NOTOC (Notification to Captain)
- [C]: Cargo bill
- [D]: Customs clearance

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under ICAO Annex 18 and EASA SPA.DG.105, the Commander must receive a written NOTOC detailing all hazardous goods, positions, and emergency drill codes.

Q2: Under which IATA Dangerous Goods Class are LITHIUM-ION BATTERIES categorized?

- [A]: Class 1
- [B]: CLASS 9 (Miscellaneous Dangerous Goods)
- [C]: Class 3
- [D]: Class 5

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Lithium batteries (UN 3480 and UN 3481) fall under Class 9: Miscellaneous Dangerous Goods.

Q3: What hazard is associated with carrying large quantities of DRY ICE (solid carbon dioxide) in an aircraft cabin?

- [A]: Explosion risk
- [B]: SUFFOCATION / ASPHYXIATION HAZARD, as dry ice sublimates into gaseous carbon dioxide, displacing oxygen in confined compartments
- [C]: Radiation
- [D]: Corrosion of steel

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Dry ice sublimates into gaseous CO₂ at normal hold temperatures, creating an invisible asphyxiation risk for ground and flight personnel.